

Major updates on 10/7

1. FPGA PCB update

Material list complete. Finalizing schematic and layout; routing has not started.

2. Alternative method: LDO → XEM8305 module

Working on schematic and routing; confirming dimensions.

Zitong’s complete component list

Updated 4k recording system FPGA PCB Board Components

FPGA + Package:

Requirements: 1. XC7A series; 2. with at least 2 GTP lanes; 3. with > 120 user I/Os; 4. with >100 in stock; 5. AMD Authorized distributors; 6. as small as possible

Final Decision:

- [XC7A35T-2CSG325I](#) (4GTPs, 150 user I/Os): JLCPCB, 122 in stock

Note: smaller package than 15*15 will not support enough user I/Os, 117 user I/Os needed for 8 chips in total, 4 chips can stimulate

LEDs:

1 for power, 1 for FPGA configuration status (DONE), different colors

Reference clock (For both FPGA & GTP):

Requirements: 1. differential reference clock 2. ~125MHz 3. peak-to-peak cycle-to-cycle jitter < 10ps; 4. <±20 ppm

Confirmed with Jiaao, we can have only one oscillator, and the clock management modules within the FPGA can take care of different clock domains.

Options:

- Abracon [AX3DCF1-125.0000T3](#): 125 MHz, 1.8 V, LVDS differential XO, 3.2 × 2.5 mm, ±25 ppm, 186 fs RMS phase jitter typ.

Need:

- 100 nF external AC-coupling capacitors;
- [TILMK1D2102LRGTR](#): 1:2 LVDS clock buffer, 1.8 V, up to 2 GHz, ultra-low additive jitter (<60 fs RMS max), VQFN-16 (3 × 3 mm), for splitting 125 MHz clock to MGTREFCLK and MRCC.

Flash:

Requirements: 1. small size 2. 1 SPI is enough 3. Vivado can recognize 4. Artix-7 supports

Workflow: MCU → JTAG → FPGA → SPI → Flash

Options:

- [ISSI IS25WP128F-JKLE](#), 128 Mbit Serial NOR Flash, 1.65–1.95 V supply, up to 166MHz 6 × 5 mm, revision control version of IS25WP128-JKLE (used by SpikeGadget), Stock: 45 in JLCPCB, 1111 expected in Mouser on 12/16/2026 (tape-and-reel version 8000+ expected next week)

Need: Level translator, NXP [NVT4858HKZ](#): 1.08–1.95 V A-side, 1.65–3.6 V B-side, auto-direction bidirectional level translator, up to 208 MHz, XQFN-16 (2.6 × 1.8 mm), suitable for FPGA 1.5 V → Flash 1.8 V, new version of SpikeGadget NVT4857UK

Note: Like the NVT4857UK used by SpikeGadgets, NVT4858 is originally designed for SD/SDIO memory interfaces rather than FPGA configuration flash. The voltage and speed requirements match, but boot/configuration behavior should still be validated for our design.



Zitong’s complete original 10/7 list

FPGA PCB schematic

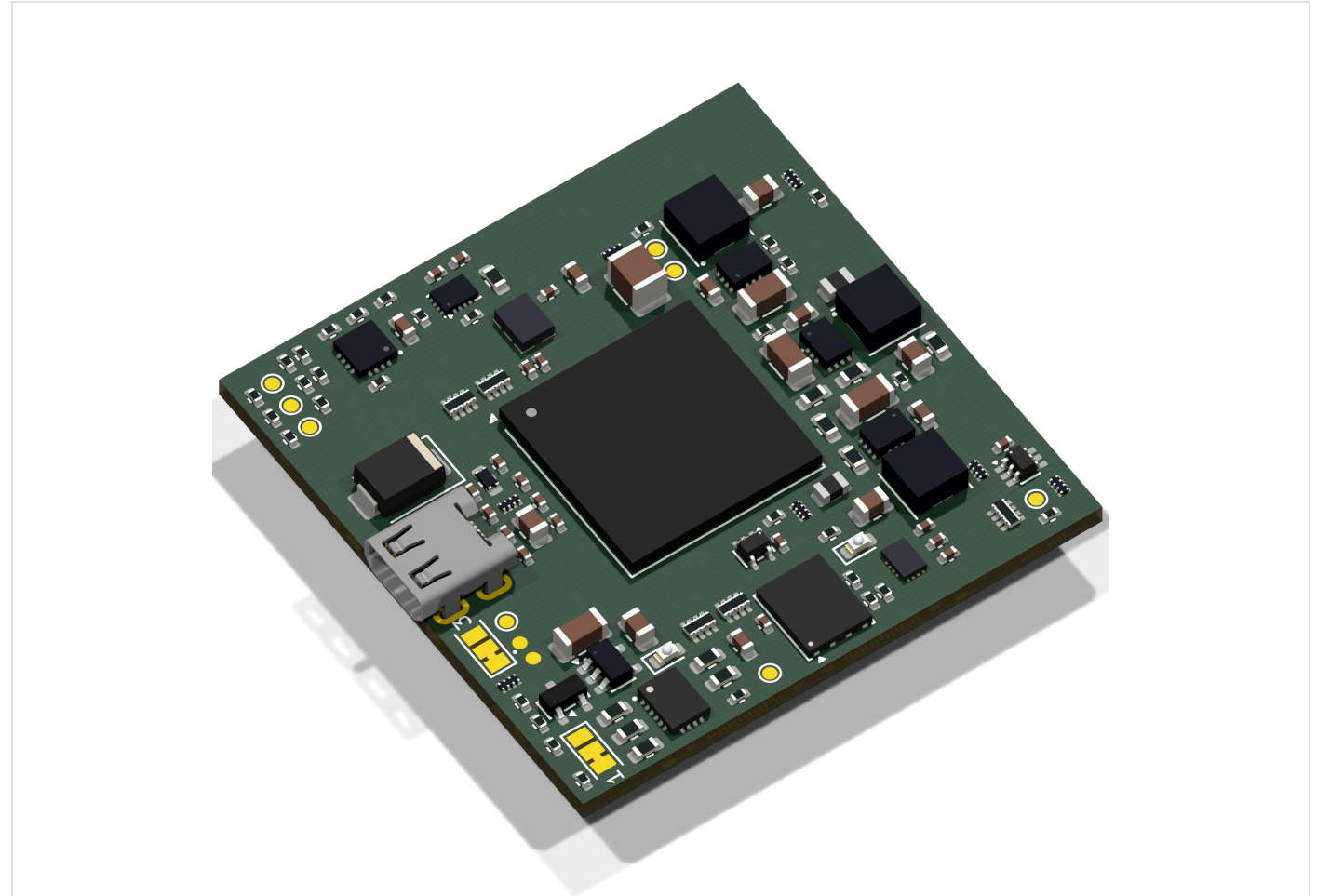
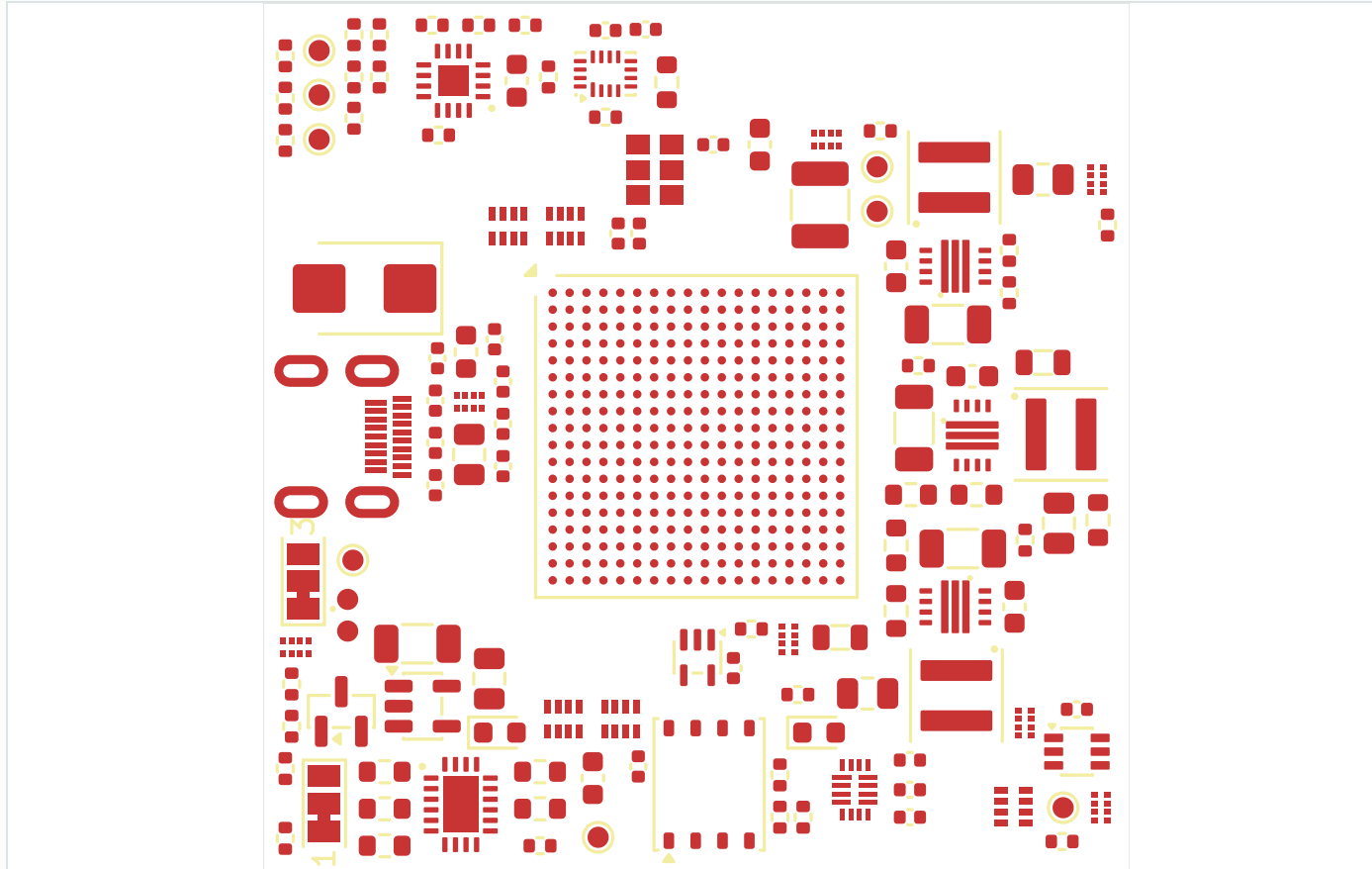


Power, startup and link



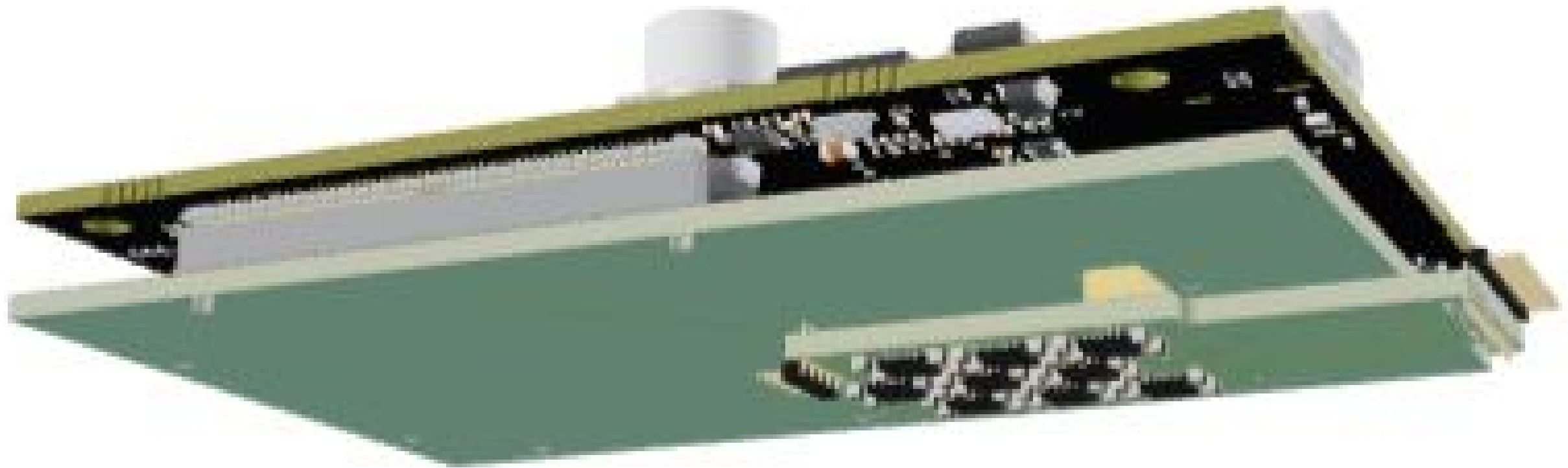
ASIC signals and connectors

FPGA PCB layout · KiCad and 3D



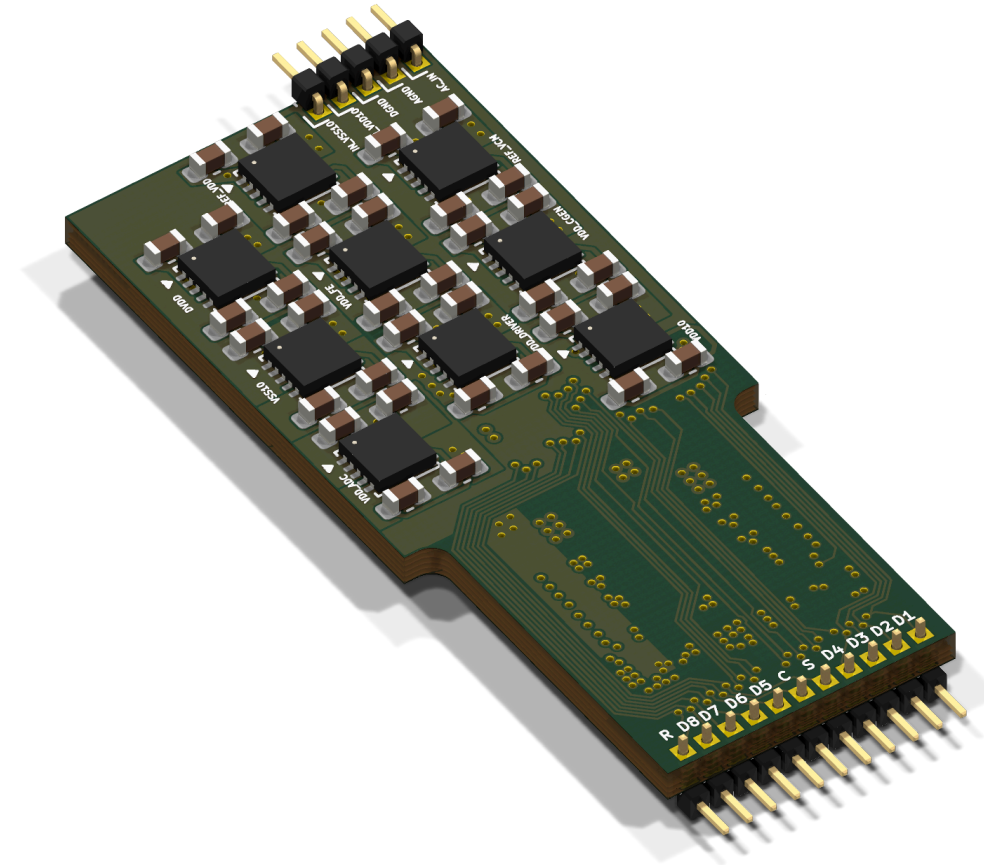
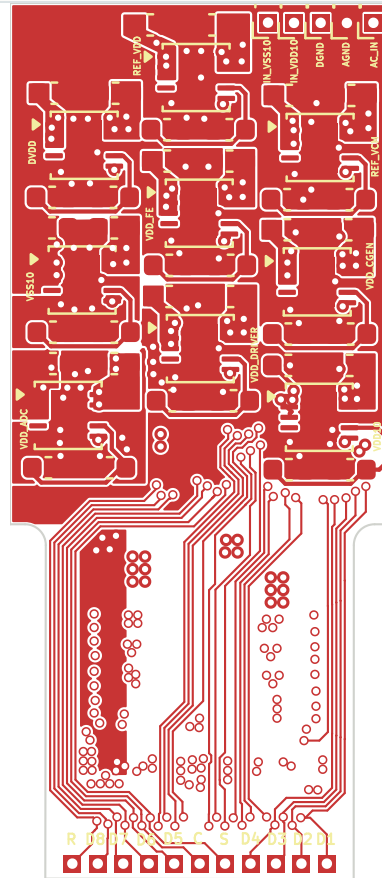
41 × 41 mm · layout in progress · routing has not started.

XEM8305, carrier and LDO



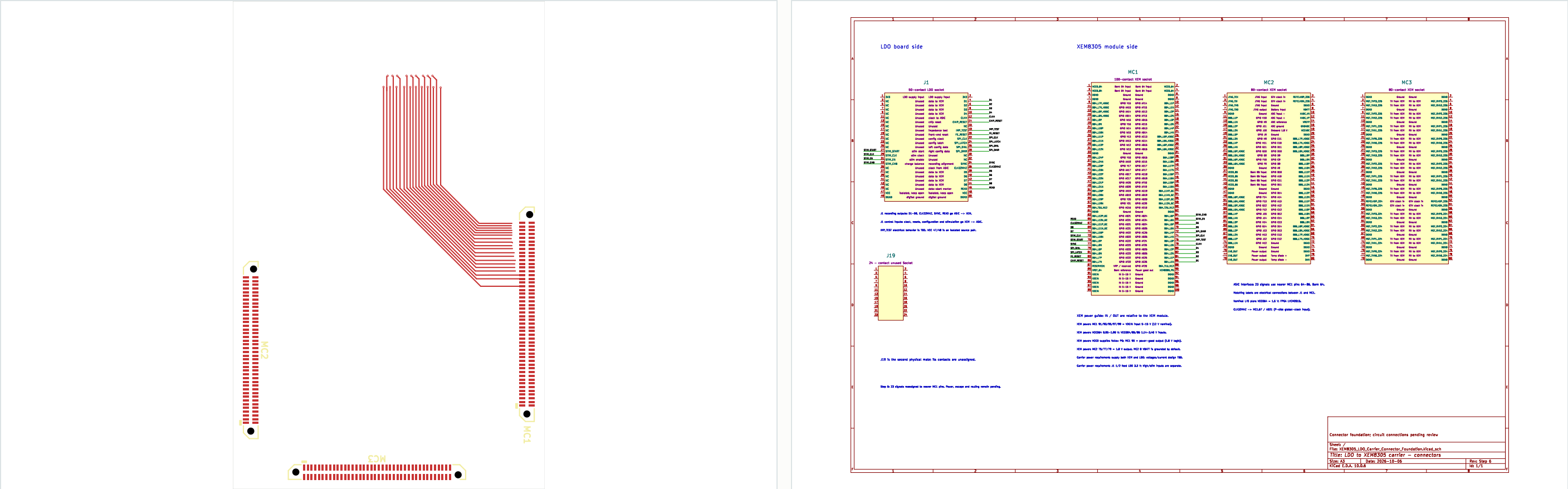
Nominal assembly; dimensions and connector fit under review.

Existing ASIC LDO board



Existing LDO design; fitted rails and connector fit under review.

Current carrier PCB - KiCad and schematic



Moving forward

1. Confirm and complete the FPGA PCB schematic.
2. Begin routing the FPGA PCB.
3. Confirm the carrier board layout, schematic and routing.